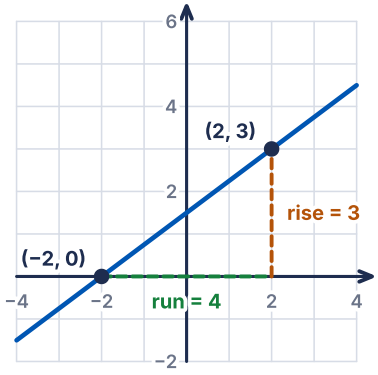
Understanding Slope

ANSWER KEY

Guided Notes

What is Slope?

Imagine you're walking up a hill. Some hills are steep, some are gentle. **Slope** tells how steep a line is.



Walk right 4, climb up 3. Slope measures that climb.

Slope is how much y changes when x goes up 1. This is the *rate of change*. It tells us two things:
1. How steep the line is
2. Which direction it goes (up or down, left to right)

We use the letter **m** to represent slope. In the equation **y = mx + b**:

m = the slope (steepness and direction)
b = the y-intercept (where the line crosses the y-axis)

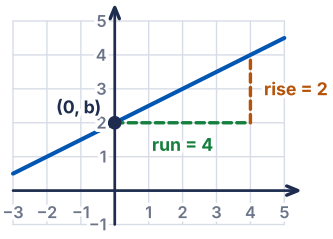
📝 Write this in your notes: "Slope = rate of change = rise over run = m"

✏ Practice: What Is Slope?

1. ★ Try it In **y = mx + b**, what does the letter **m** stand for?

- ● **the slope**
- ○ the y-intercept
- ○ the x-value

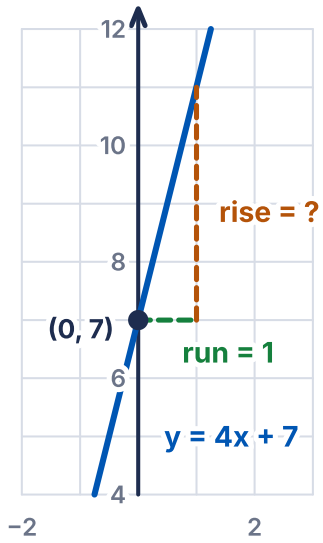
How to do it: In y = mx + b, **m** is the slope. (b is the y-intercept.)



2. ★★ On level What is the slope of **y = 4x + 7**? Write just the number.

Answer: 4

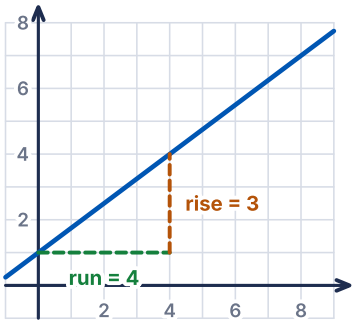
How to do it: Match it to y = mx + b. The number in front of x is m, so m = **4**. (7 is b, the y-intercept.)



3. ★★★ Challenge This hill goes right 4 and up 3. If you walk right 8 on the same hill, how far up do you climb?

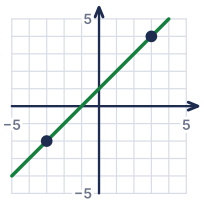
Answer: 6

How to do it: Right 8 is 2 groups of right 4. Each group climbs 3. So you climb 3 + 3 = **6**. The slope stays the same: $\frac{3}{4} = \frac{6}{8}$.



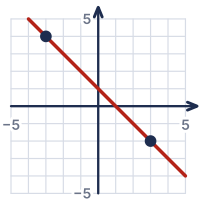
Four Types of Slope

There are exactly **four types** of slope:



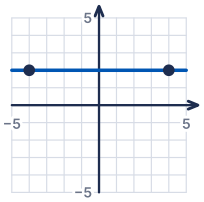
Positive ↗

m > 0



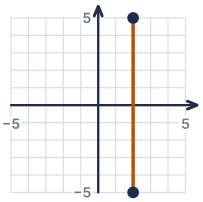
Negative ↘

m < 0



Zero →

m = 0



Undefined Slope ↕

Vertical line.

m = undefined

Positive slope: As x increases, y also increases. Think of walking *uphill*.

Negative slope: As x increases, y decreases. Think of walking *downhill*.

Zero slope: A perfectly flat, horizontal line. No rise at all — the y-value never changes. Think of walking on *flat ground*.

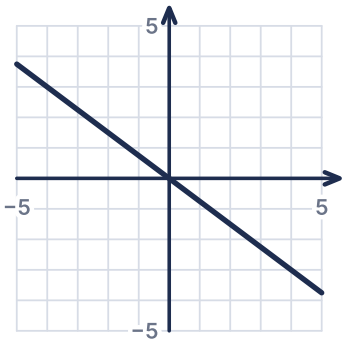
Undefined slope: A vertical line. The run is 0, and we can't divide by zero! Think of a *cliff wall* — you can't walk up that.

📝 Memory trick: **"Zero slope = Zero (flat), Undefined = Up and Down (vertical)"**

✏ Practice: Four Types of Slope

1. ★ Try it What type of slope does this line have?

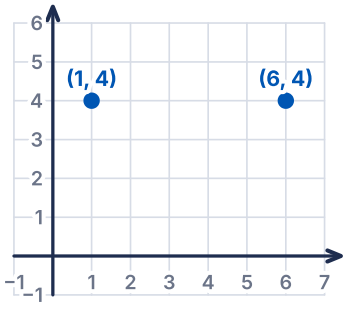
- ○ Positive
- ● **Negative**
- ○ Zero
- ○ Undefined



How to do it: Reading left to right, the line goes **down** (downhill). So the slope is **negative**.

2. ★★ On level A line goes through **(1, 4)** and **(6, 4)**. What type of slope does it have?

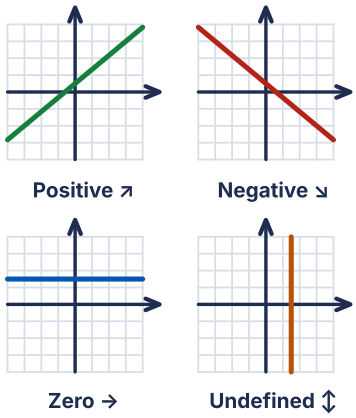
- ○ Positive
- ○ Negative
- ● **Zero**
- ○ Undefined



How to do it: Both points have y = 4, so the line never goes up or down. It is flat (horizontal). The slope is **zero**.

3. ★★★ Challenge Which equation has a **negative** slope?

- ○ y = 3x − 5
- ● **y = −2x + 4**
- ○ y = x + 1
- ○ y = 5x



How to do it: The slope is the number in front of x: 3, −2, 1 and 5. Only −2 is negative, so **y = −2x + 4** goes downhill.

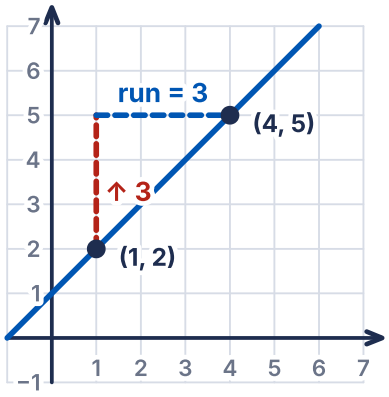
Finding Slope from a Graph: Rise Over Run

To find slope from a graph, pick **any two points** on the line, then count:

$$\text{Slope} = \frac{\text{rise (vertical change)}}{\text{run (horizontal change)}}$$

Rise = how far you move **up or down** (count vertical grid squares)
Run = how far you move **left or right** (count horizontal grid squares)

Example 1: Line through (1, 2) and (4, 5)



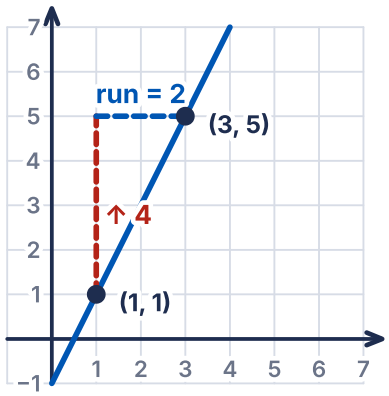
Starting at **(1, 2)**:

→ Count **UP 3** squares (rise = 3)
→ Count **RIGHT 3** squares to reach (4, 5) (run = 3)

> Slope = $\frac{\text{rise}}{\text{run}} = \frac{3}{3} = 1$

Example 2: A steeper line

Line through **(1, 1)** and **(3, 5)**.

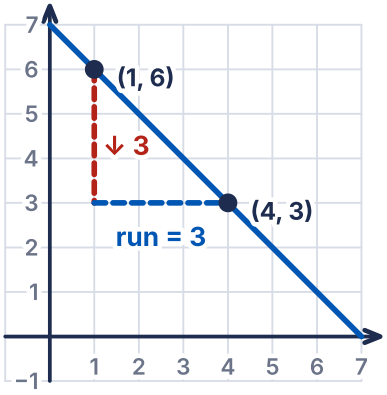


→ **Rise = 4** (count up from 1 to 5)
→ **Run = 2** (count right from 1 to 3)

> Slope = $\frac{4}{2} = 2$

Example 3: Negative slope (going downhill)

Line through **(1, 6)** and **(4, 3)**.



→ **Rise = −3** (count DOWN from 6 to 3 — negative because you go down!)
→ **Run = 3** (count right from 1 to 4)

> Slope = $\frac{-3}{3}$ = **−1**

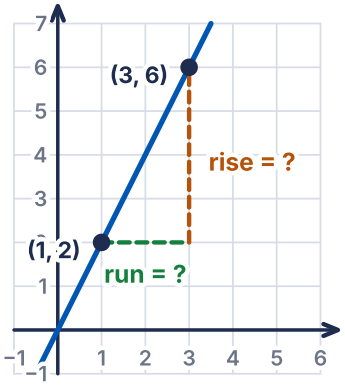
> 📝 If you count **down**, the rise is **negative**. That's what makes the slope negative!

✏ Practice: Rise Over Run

1. ★ Try it Count the rise and the run from one dot to the other. What is the slope?

Answer: 2

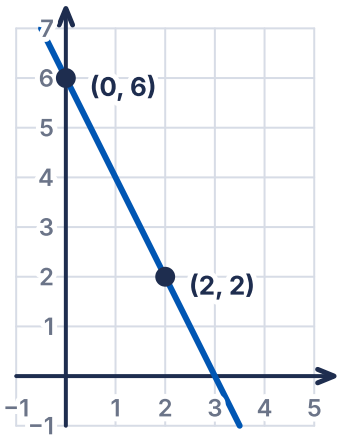
> **How to do it:** Rise: up 4 (from y = 2 to y = 6). Run: right 2 (from x = 1 to x = 3). Slope = $\frac{4}{2}$ = **2**.



2. ★★ On level Find the slope of this line. Use the two dots.

Answer: −2

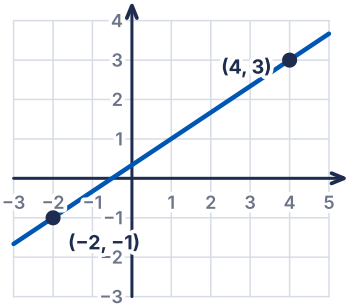
How to do it: From (0, 6), count **down** 4 to y = 2: rise = −4. Then right 2: run = 2. Slope = $\frac{-4}{2}$ = **−2**.



3. ★★★ Challenge Find the slope of this line. Write it as a fraction in simplest form, like $\frac{3}{4}$.

Answer: $\frac{2}{3}$

How to do it: Rise: up 4 (from −1 to 3). Run: right 6 (from −2 to 4). Slope = $\frac{4}{6}$. Divide top and bottom by 2: $\frac{2}{3}$.



The Slope Formula

When you have two points but no graph, use the **slope formula**:

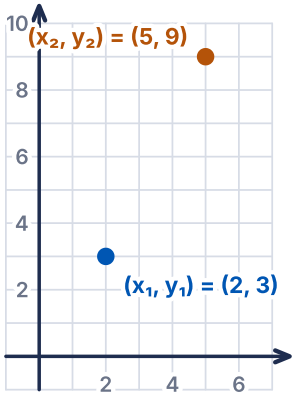
$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

(x_1, y_1) is the first point
(x_2, y_2) is the second point
The **numerator** (top) is the rise. The **denominator** (bottom) is the run.

Example 1: Find the slope of the line through (2, 3) and (5, 9)

Step 1: Label your points

$(x_1, y_1) = (2, 3)$
$(x_2, y_2) = (5, 9)$



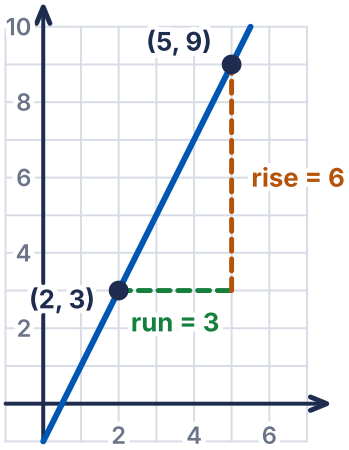
Example 1 (continued)

Step 2: Substitute into the formula

$$m = \frac{9 - 3}{5 - 2}$$

Step 3: Simplify

$$m = \frac{6}{3} = 2$$



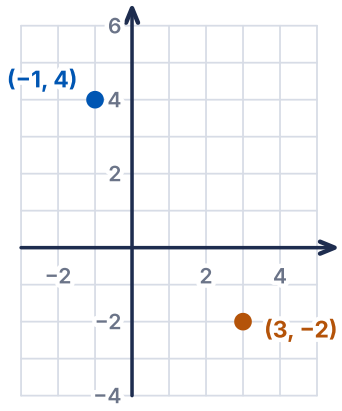
rise ÷ run = 6 ÷ 3 = 2

📝 The slope is **2**, which means for every 1 unit you move right, you move up 2 units.

Example 2: (−1, 4) and (3, −2)

Step 1: Label your points

$(x_1, y_1) = (-1, 4)$
$(x_2, y_2) = (3, -2)$



Step 2: Substitute

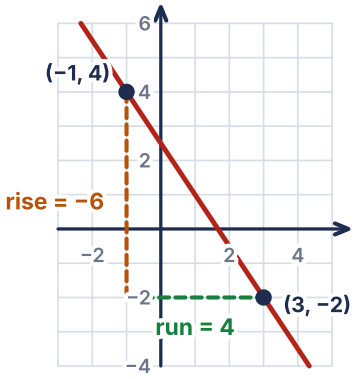
$$m = \frac{-2 - 4}{3 - (-1)}$$

Example 2 (continued)

Step 3: Simplify carefully — watch the signs!

$$m = \frac{-6}{4} = \frac{-3}{2}$$

> **Important:** Always simplify your fraction! $-\frac{6}{4}$ reduces to $-\frac{3}{2}$.
> The negative sign means the line goes *downhill* (left to right).



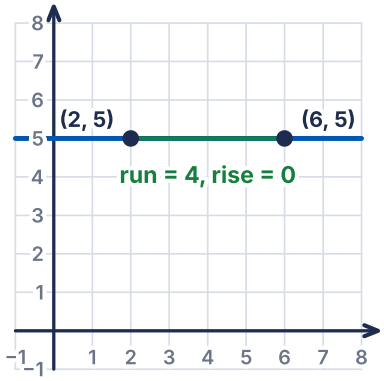
Going DOWN 6 while going right 4: $m = -6 \div 4 = -\frac{3}{2}$

> ⚠️ **Common mistake:** 3 − (−1) = 3 + 1 = 4, NOT 3 − 1 = 2. Subtracting a negative = adding!

Example 3: (2, 5) and (6, 5)

$$m = \frac{5 - 5}{6 - 2} = \frac{0}{4} = 0$$

Zero in the numerator → slope = 0 → horizontal line!
Both y-values are the same, so there's no vertical change.

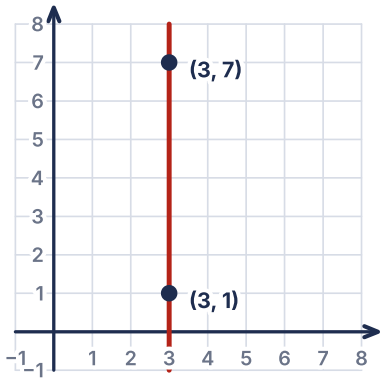


Flat line: rise = 0, so m = 0

Example 4: (3, 1) and (3, 7)

$$m = \frac{7 - 1}{3 - 3} = \frac{6}{0} = \text{undefined}$$

Zero in the denominator → slope is undefined → vertical line!
Both x-values are the same. Division by zero is undefined.



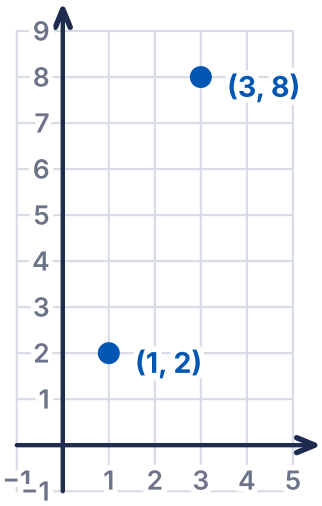
Vertical line: run = 0, and we cannot divide by 0

✏ Practice: The Slope Formula

1. ★ Try it Use the slope formula to find the slope through **(1, 2)** and **(3, 8)**.

Answer: 3

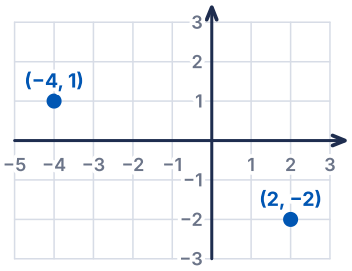
How to do it: $m = \frac{8 - 2}{3 - 1} = \frac{6}{2} = 3$.



2. ★★ On level Find the slope through **(−4, 1)** and **(2, −2)**. Write a fraction in simplest form.

Answer: $-\frac{1}{2}$

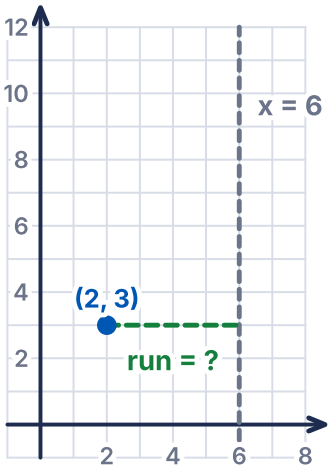
How to do it: $m = \frac{-2 - 1}{2 - (-4)} = \frac{-3}{6}$. Simplify: $-\frac{1}{2}$.



3. ★★★ Challenge A line goes through **(2, 3)** and **(6, y)**. Its slope is 2. What is y?

Answer: 11

How to do it: Run: 6 − 2 = 4. Slope 2 means rise ÷ 4 = 2, so the rise is 8. y = 3 + 8 = **11**. Check: $\frac{11 - 3}{6 - 2} = \frac{8}{4} = 2$ ✓



(6, y) is somewhere on the dashed line x = 6.

Comprehensive Practice

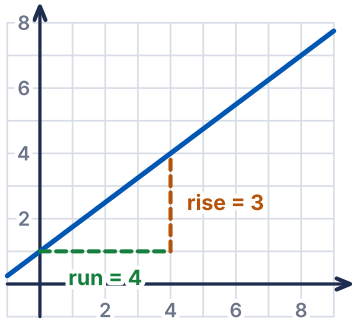
Mixed practice from every section: what slope means, the four types, rise over run, and the slope formula. If an answer is a fraction, write it like $\frac{3}{4}$ or $-\frac{1}{2}$.

✏ Comprehensive Practice All sections

1. ★ Try it What does slope measure?

- ● **How steep a line is**
- ○ Where a line crosses the y-axis
- ○ How long a line is

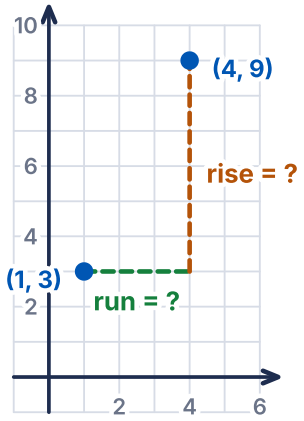
How to do it: Slope measures how **steep** a line is (and which way it goes). Where the line crosses the y-axis is b, the y-intercept.



2. ★ Try it Find the slope through (1, 3) and (4, 9).

Answer: 2

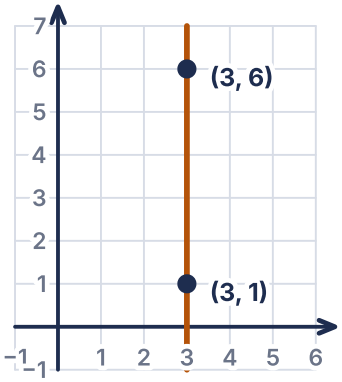
How to do it: $m = \frac{9 - 3}{4 - 1} = \frac{6}{3} = 2$.



3. ★★★ Challenge Jay says, "A vertical line has a slope of 0." Is Jay right?

- ○ Yes, a vertical line has slope 0
- ● **No, a vertical line has an undefined slope**
- ○ No, a vertical line has a negative slope

How to do it: No. A vertical line has run = 0, and we can't divide by 0. So its slope is **undefined**. A *horizontal* (flat) line is the one with slope 0.

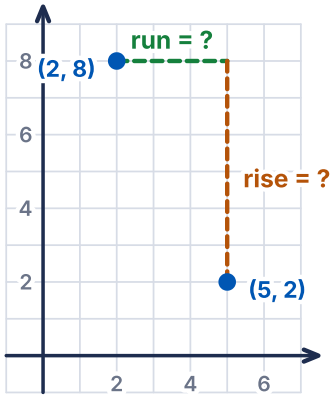


A vertical line

4. ★★ On level Find the slope through (2, 8) and (5, 2).

Answer: −2

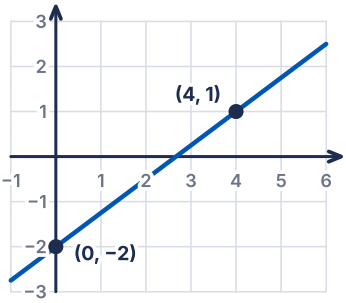
How to do it: $m = \frac{2 - 8}{5 - 2} = \frac{-6}{3} = -2$.



5. ★★ On level What is the slope of this line? Write a fraction, like $\frac{1}{2}$.

Answer: $\frac{3}{4}$

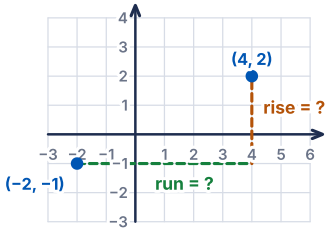
How to do it: Rise: up 3 (from −2 to 1). Run: right 4 (from 0 to 4). Slope = $\frac{3}{4}$.



6. ★★★ Challenge Find the slope through (−2, −1) and (4, 2). If the answer is a fraction, write it as $\frac{a}{b}$ (like $\frac{3}{4}$).

Answer: $\frac{1}{2}$

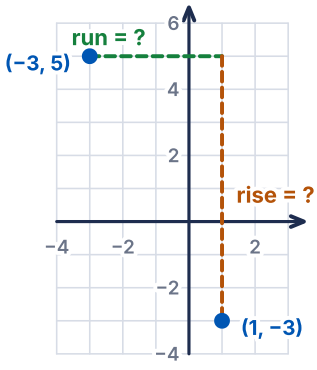
How to do it: $m = \frac{2 - (-1)}{4 - (-2)} = \frac{3}{6}$. Simplify: $\frac{1}{2}$.



7. ★★ On level Find the slope through (−3, 5) and (1, −3).

Answer: −2

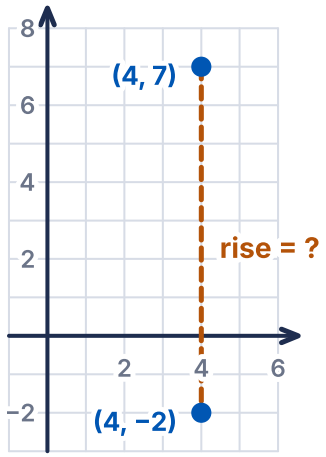
How to do it: $m = \frac{-3 - 5}{1 - (-3)} = \frac{-8}{4} = -2$.



8. ★★ On level Find the slope through (4, 7) and (4, −2). Write "undefined" if the slope is undefined.

Answer: undefined

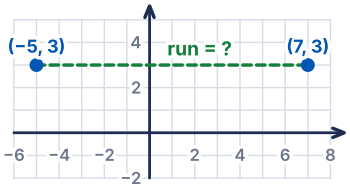
How to do it: $m = \frac{-2 - 7}{4 - 4} = \frac{-9}{0}$. We can't divide by 0, so the slope is **undefined** (a vertical line).



9. ★ Try it Find the slope through (−5, 3) and (7, 3).

Answer: 0

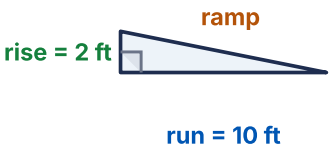
How to do it: $m = \frac{3 - 3}{7 - (-5)} = \frac{0}{12} = 0$ (a horizontal line).



10. ★★★ Challenge A ramp rises 2 feet over a run of 10 feet. What is its slope? Write a fraction in simplest form.

Answer: $\frac{1}{5}$

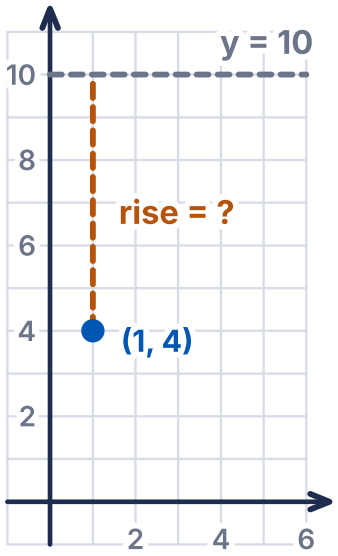
How to do it: slope $= \frac{\text{rise}}{\text{run}} = \frac{2}{10}$. Divide top and bottom by 2: $\frac{1}{5}$.



11. ★★★ Challenge A line goes through **(1, 4)** and **(x, 10)**. Its slope is 3. What is x?

Answer: 3

How to do it: Rise: 10 − 4 = 6. Slope 3 means 6 ÷ run = 3, so the run is 2. x = 1 + 2 = **3**. Check: $\frac{10 - 4}{3 - 1} = \frac{6}{2} = 3$ ✓



(x, 10) is somewhere on the dashed line y = 10.